

College of Industrial Technology
King Mongkut's University of Technology North Bangkok

Seat No.

Final Examination of Semester 1

Year: 2024

Subject: 031001113 WAVE MOTION AND SOUND

Section: 15-18

Date: 2 October 2024

Time: 8.00-10.00

Name: _____ ID: _____ Class: _____

Directions: The test is designed to measure your comprehension. The test is divided into 2 sections. There will be 9 pages (including this page) and they are worth 35 points.

- This exam paper contains no errors. If a suspected error is found, it is the student's discretion to correct it.
- Answer the questions on this test paper.
- Books, documents, and lecture notes are not allowed.
- You must be in the room for one hour after the exam is started and, while taking the exam, you cannot go out except in an emergency case.
- Before leaving, make sure you do not bring this test outside.
- Do not use any electronic communication device.
- Calculators cannot be used in this test.

Now begin the test.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

This exam paper is part of a set used to assess student abilities as follows:

		Section 1	Section 2
CLO 4	Do experiment and explain the reflection of light, plan mirror, curved mirror. Including calculating various related quantities	1,2,3,7,13,14, 15,16	1.1
CLO 5	Do experiment and explain the refraction of light, lens. Including calculating various related quantities	4,5,6,7,8,17, 18,19	1.2
CLO 6	Explain wave phenomena of light, interference, diffraction, polarization. observe and explain various properties of waves, as well as calculate speed, frequency, wavelength, and related quantities.	9,10,11,12,20, 21,22,23	2.1,2.2

Useful formulas

$$\frac{1}{d_o} + \frac{1}{d_i} = \frac{1}{f} \quad M = \frac{h_i}{h_o} = -\frac{d_i}{d_o} \quad n = \frac{c}{v} \quad \lambda_1 n_1 = \lambda_2 n_2 \quad n_1 \sin \theta_1 = n_2 \sin \theta_2$$

$$d' = d \left(\frac{n_2}{n_1} \right) \quad \sin \theta_c = \frac{n_2}{n_1} \quad P = \frac{1}{f} \quad \frac{1}{f} = (n-1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right) \quad M = \frac{\theta'}{\theta}$$

$$M = 1 + \frac{N}{f} \quad d \sin \theta = m\lambda, \quad m = 0, 1, 2, 3, \dots \quad d \sin \theta = \left(m + \frac{1}{2}\right)\lambda, \quad m = 0, 1, 2, 3, \dots$$

$$D \sin \theta = m\lambda, \quad m = 1, 2, 3, \dots \quad d \sin \theta = m\lambda, \quad m = 0, 1, 2, 3, \dots \quad I = I_{\max} \cos^2 \theta$$

$$c = 3.00 \times 10^8 \text{ m/s}$$

Session 1: Multiple-choice Question (23 questions) (1 mark / question)

Instruction: mark X for a correct choice of each question on the answer sheet provided.

Answer sheet											
	A	B	C	D	E		A	B	C	D	E
1						13					
2						14					
3						15					
4						16					
5						17					
6						18					
7						19					
8						20					
9						21					
10						22					
11						23					
12											

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6. A plano-convex glass ($n = 1.5$) lens has a curved side whose radius is 50 cm. If the image size is to be the same as the object size, the object should be placed at a distance from the lens of:

- A) 50 cm B) 100 cm C) 200 cm D) 340 cm E) 400 cm

7. Of the possible statements that parallel rays are brought to a focus by a

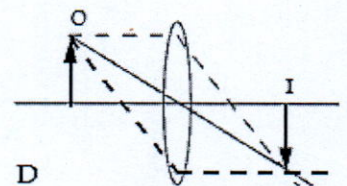
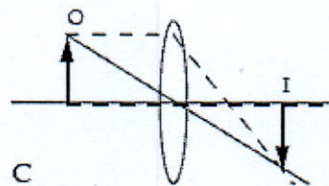
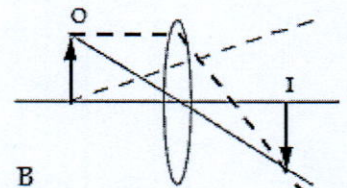
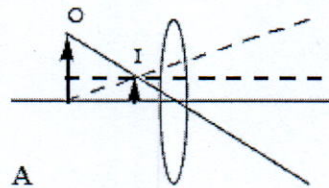
- 1) concave mirror, 2) convex mirror,
3) converging lens, 4) diverging lens,

the correct answer is

- A) 1 and 2. B) 3 and 4. C) 1 and 3. D) 2 and 3. E) 2 and 4.

8. Which ray diagrams are correct? The three rays in each diagram are distinguished by different types of lines. Hint: Check the behavior of each ray as it interacts with lens.

- A) B and C
B) C and D
C) D and E
D) C and E
E) A, C, and D



9. Which of the following types of electromagnetic radiation travels at the greatest speed in vacuum?

- A) Radio waves B) Visible light
C) X rays D) Gamma rays
E) All of these travel at the same speed

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Question 13-23: Read each statement below carefully.

Place an X on the choice A in the answer sheet if you think the statement is TRUE.

Place an X on the choice E in the answer sheet if you think the statement is FALSE.

13. The images form by the convex mirrors are always smaller than objects.
14. When you stand in front of a plane mirror, your image is virtual, and the same size as you.
15. As an object is moved from the center of curvature of a concave mirror toward its focal point, its image remains real and becomes larger.
16. The focal length of convex mirror is negative.
17. The unit of index of refraction is m/s.
18. The index of refraction of a certain glass is 1.50. The sine of the critical angle ($\sin \theta_c$) for total internal reflection at a glass-air interface is 0.5.
19. When light travels from one medium into a different medium with a different index of refraction, the speed stays the same.
20. In a Young's double-slit experiment, the slit separation is doubled. To maintain the same fringe spacing on the screen, the screen-to-slit distance L must be changed to $2L$.
21. A parallel beam of monochromatic light is incident on a single slit of width 0.2 mm. The light passing through the slit falls on a screen 2 m away. As the slit width is decreased, the width of the pattern on the screen remains the same.
22. The widths between the bright lines produced by monochromatic light falling on a diffraction grating can be reduced by decreasing the distance between the grating and the screen.
23. As unpolarized light is passed through a linear polarizer, its intensity will be reduced to 50%.

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Session 2: Step-by-step explanation is required for this section.

(4 questions) (3 marks/question)

1.1 Take a concave mirror to catch the sunlight. found that the light reflected from the mirror will converge to a point 15 cm in front of the mirror. Find how far an object must be placed from the mirror to produce a real image with a magnification of 3 times.

(3 marks)

1.2 A far-sighted student has a near point of 0.75 m. Calculate the focal length (in cm) of the glasses needed so the near point will be normal (25 cm).

(3 marks)

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2.1 Light of wavelength 600 nm is incident on double slits having a width of 0.300 mm. The viewing screen is 2.00 m from the slit. Find the width of the central bright fringe.
(3 marks)

2.2 The 600-nm light is incident on a diffraction grating with a ruling separation of $d = 2.4 \times 10^{-6}$ m. Determine the angle of the second-order maximum.
(3 marks)

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